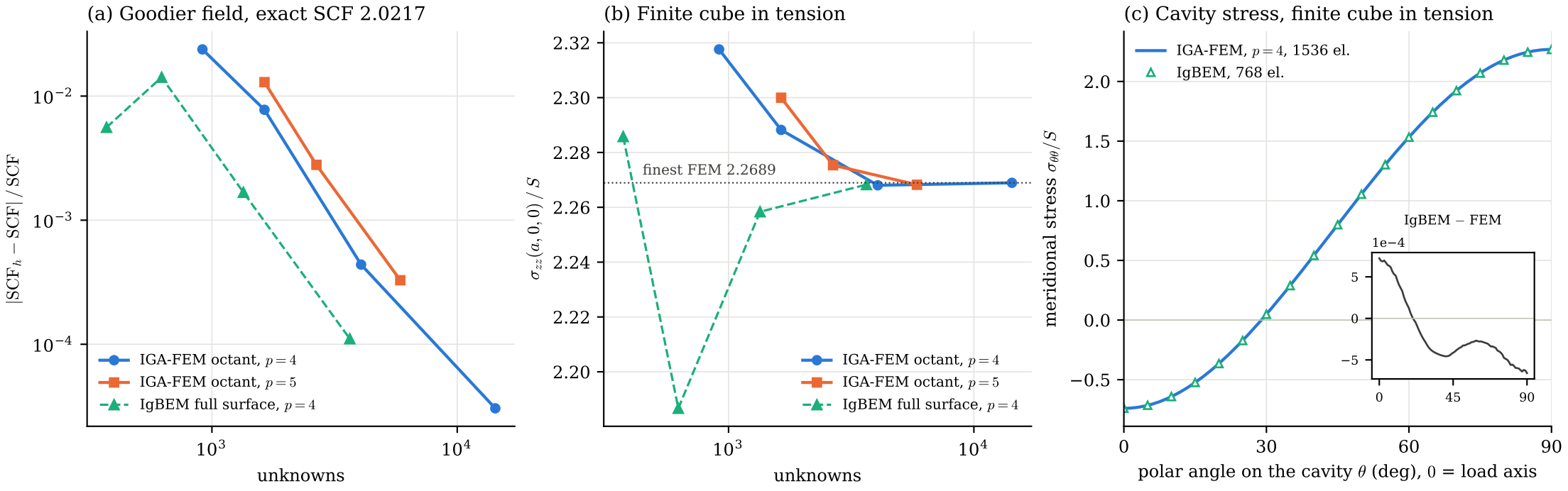
(a) Goodier field, exact SCF 2.0217
$|\mathrm{SCF}_h - \mathrm{SCF}| / \mathrm{SCF}$
10^{-2}
10^{-3}
10^{-4}
10^{3}
10^{4}
unknowns
IGA-FEM octant, $p = 4$
IGA-FEM octant, $p = 5$
IgBEM full surface, $p = 4$
(b) Finite cube in tension
$\sigma_{zz}(a, 0, 0) / S$
2.32
2.30
2.28
2.26
2.24
2.22
2.20
finest FEM 2.2689
10^{3}
10^{4}
unknowns
IGA-FEM octant, $p = 4$
IGA-FEM octant, $p = 5$
IgBEM full surface, $p = 4$
(c) Cavity stress, finite cube in tension
meridional stress $\sigma_{\theta\theta}/S$
IGA-FEM, $p = 4$, 1536 el.
IgBEM, 768 el.
2.0
1.5
1.0
0.5
0.0
−0.5
IgBEM − FEM
1e−4
5
0
−5
0
45
90
0
30
60
90
polar angle on the cavity θ (deg), 0 = load axis